

PIYUSH CHHALLARE

Semiconductor engineering graduate student | Experimental physicist

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EDUCATION

Northeastern University, Boston, MA

2026 - present

- M.S. Semiconductor Engineering, Devices & Nanosystems concentration; thesis track.

Purdue University, West Lafayette, IN

2022 - 2026

- B.S. Physics; Certificate in Quantum Information Science and Technology.

RESEARCH EXPERIENCE

Quantum Spin Lab, Purdue University

2024 - 2026

- Grew single crystals of triangular-lattice quantum materials, including TiYbS_2 ; contributed to cryogenic magnetization, heat-capacity, and diffraction studies.
- Analyzed field- and temperature-dependent measurements in Python; co-authored a preprint on the low-temperature thermodynamics of TiYbS_2 .

Summer Undergraduate Research Fellow, Purdue University

Summer 2024

- Revisited flux-growth conditions and thermal profile for TiYbS_2 ; helped obtain single crystals after earlier unsuccessful growth attempts.

Physics Data Mine, Purdue University

2023

- Used Python and Monte Carlo methods to study simulated top-quark decay data and select particle signatures.

PUBLICATION & PRESENTATION

Co-author, "Low Temperature Thermodynamics of $S_{\text{eff}} = 1/2$ Triangular Lattice Quantum Spin Liquid Candidate TiYbS_2 ," arXiv:2609.11881 (2026).

Poster, "Exploring Low Temperature Magnetism in a Novel Rare-Earth Trihalide," APS Global Physics Summit (2026).

SELECTED SKILLS

Experimental: flux crystal growth, sample preparation, MPMS, PPMS, AC/DC susceptibility, heat capacity, X-ray diffraction, Hall measurements.

Analysis & engineering: Python, NumPy, pandas, SciPy, Matplotlib, curve fitting, semiconductor device physics; currently studying analog IC design, MEMS, and computer architecture.